

NAME:JEEVITHA S

SUB:DATABASE MANAGEMENT SYSTEMS

Application Name: Food Delivery Management System

1.Purpose:

The Food Delivery Management System is a database-driven application that allows customers to browse restaurants and food items, place orders, make payments, and track their deliveries. It also helps restaurants manage their menus and orders, while delivery partners manage assigned deliveries.

2.List of Required Tables:

A practical Food Delivery System can be represented using **8 main tables:**

1. **Customer**
2. **Restaurant**
3. **Menu_Item**
4. **Order**
5. **Order_Item**
6. **Payment**
7. **Delivery_Partner**
8. **Delivery**

3. Tables and Columns/Attributes

Table 1: Customer

Column	Datatype	Key	Description
Customer_ID	INT	Primary key	Unique customer ID
Name	VARCHAR	-	Customer's name
Phone	VARCHAR	-	Contact number
Email	VARCHAR	-	Customer email
Address	VARCHAR	-	Delivery address

Purpose: Stores information about customers who use the food delivery application.

Table 2: Restaurant

Column	Datatype	key	Description
Restaurant_ID	INT	Primary key	Unique restaurant ID
Restaurant_name	VARCHAR	-	Name of restaurant

Phone	VARCHAR	-	Restaurant contact number
Address	VARCHAR	-	Restaurant location
Rating	DECIMAL	-	Restaurant rating

Purpose: Stores information about restaurants available on the platform.

Table 3: Menu_Item

Column	Datatype	Key	Description
Item_ID	INT	Primary key	Unique food item ID
Restaurant_ID	INT	Foreign key	Restaurant offering the item
Item_name	VARCHAR	-	Name of food item
Category	VARCHAR	-	Food category
Price	DECIMAL	-	Price of item
Availability	BOOLEAN	-	Whether item is currently available

Purpose: Stores the food items offered by each restaurant.

Relationship: One restaurant can have many menu items.

Table 4: Order

Column	Datatype	Key	Description
Order_ID	INT	Primary key	Unique order ID
Customer_ID	INT	Foreign key	Customer placing the order
Restaurant_ID	INT	Foreign key	Restaurant receiving the order
Order_Data	DATETIME	-	Date and time of order
Total_amount	DECIMAL	-	Total order value
Order_status	VARCHAR	-	Pending,preparing,delivered,vancelled, etc..

Purpose: Stores the main information about every order placed by customers.

Table 5: Order_Item

Column	Datatype	Key	Description
Order_item_ID	INT	Primary key	Unique order-item ID
Order_ID	INT	Foreign key	Related order
Item_ID	INT	Foreign key	Food item ordered
Quantity	INT	-	Number of items ordered
Unit_price	DECIMAL	-	Price of item at time of order

Purpose: Connects orders with the individual food items purchased.

Table 6: Payment

Column	Datatype	Key	Description
Payment_Id	INT	Primary key	Unique payment ID
Order_id	INT	Foreign key	Related order
Payment_method	VARCHAR	-	UPI,Card,cash,etc.
Payment_status	VARCHAR	-	Pending ,successful,failed
Payment_date	DATETIME	-	Date and time of payment
Amount	DECIMAL	-	Amount paid

Purpose: Records payment information associated with an order.

Table 7: Delivery_Partner

Column	Datatype	Key	Description
Partner_ID	INT	Primary key	Unique delivery partner name ID
Name	VARCHAR	-	Delivery partner name
Phone	VARCHAR	-	Contact name
Vehicle_type	VARCHAR	-	Bike ,scooter,etc.
Partner_status	VARCHAR	-	Available,busy ,offline

Purpose: Stores information about delivery partners who deliver customer orders.

Table 8: Delivery

Column	Datatype	key	Description
Delivery_ID	INT	Primary key	Unique delivery ID
Order_ID	INT	Foreign key	Order being delivered
Partner_ID	INT	Foreign key	Assigned delivery partner
Pickup_Time	DATETIME	-	Time order was picked up
Delivery_Time	DATETIME	-	Time order was delivered
Delivery_Status	VARCHAR	-	Assigned, Picked Up, Delivered

Purpose: Tracks the delivery of an order from the restaurant to the customer.

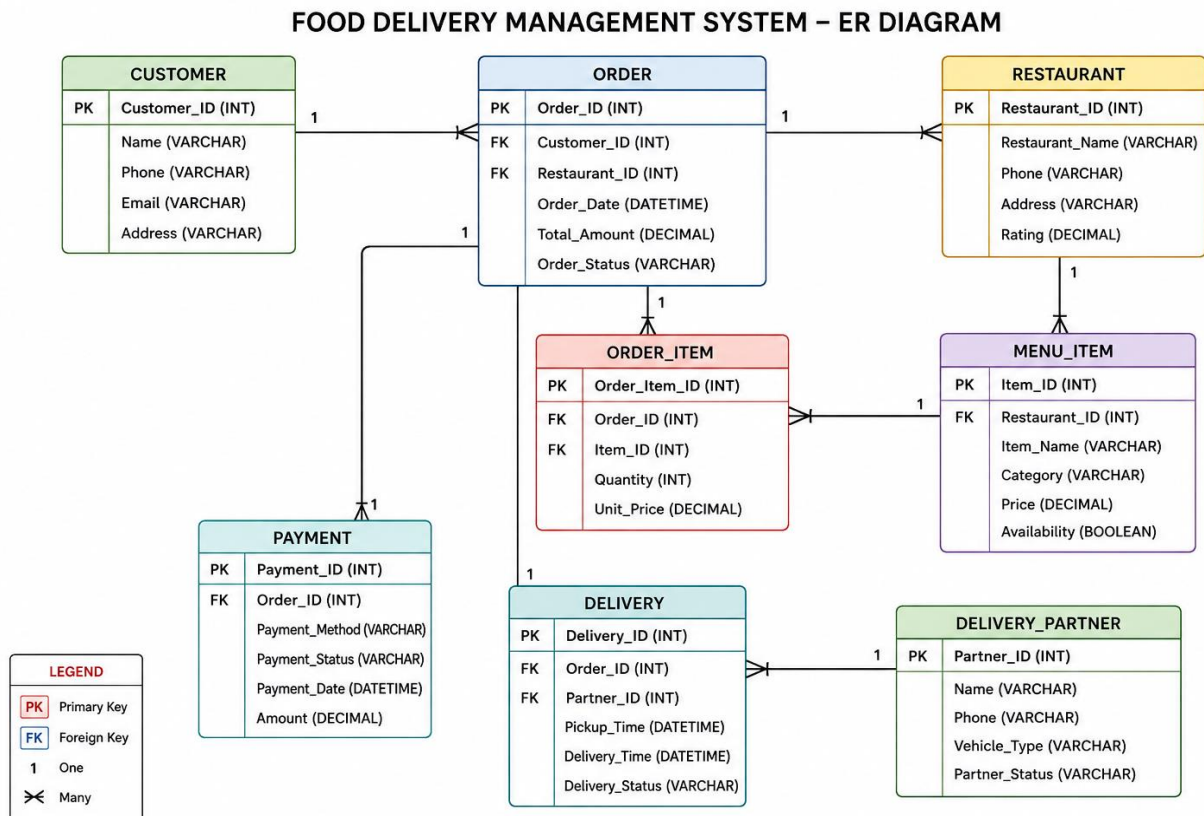
4. Relationships Between Tables

The main relationships are:

Relationship	Type	Explanation
Customer → Order	1:many	One customer can place many orders
Restaurant → Menu_Item	1;many	One restaurant can offer many food items
Restaurant → Order	1:many	One restaurant can receive many orders
Order → Order_Item	1:many	One order can contain many items
Menu_Item → Order_Item	1:many	One food item can appear in many orders

Order → Payment	1:1	An order has its associated payment
Order → Delivery	1:1	An order has its associated delivery
Delivery_Partner → Delivery	1:Many	One delivery partner can handle many deliveries

5. Overall Database Structure



The complete flow of the application can be represented as:

Customer → Restaurant/Menu → Order → Order Items → Payment → Delivery Partner → Delivery

For example:

A customer selects food items from a restaurant's menu and places an order. The order is stored in the **Order** table, while its individual food items are stored in **Order_Item**. The customer makes a payment recorded in **Payment**, and a delivery partner is assigned through **Delivery**.

6. Why These Tables Are Necessary

These **8 tables are sufficient** to represent the core operations of a real-world food delivery application without unnecessarily creating tables.

- **Customer** → Who orders?
- **Restaurant** → Where does the food come from?
- **Menu_Item** → What food is available?
- **Order** → What order was placed?
- **Order_Item** → Which items and quantities are in the order?
- **Payment** → How was the order paid for?
- **Delivery_Partner** → Who delivers it?
- **Delivery** → What is the delivery status?

This structure also reduces data duplication and clearly demonstrates the use of **primary keys, foreign keys, one-to-many relationships, and one-to-one relationships** in a real-world database.